

For the use of Registered Medical Practitioner or a Hospital or a Laboratory only

Seroflo Inhaler (CFC FREE)

(Salmeterol & Fluticasone Propionate Inhaler)

Composition:

Seroflo-50 Inhaler (CFC FREE)

Each actuation delivers:

Salmeterol (as Salmeterol Xinafoate) Ph. Eur.....25 mcg

Fluticasone Propionate BP50 mcg

Suspended in Propellant 134a.....q.s.

Seroflo-125 Inhaler (CFC FREE)

Each actuation delivers:

Salmeterol (as Salmeterol Xinafoate) Ph. Eur.....25 mcg

Fluticasone Propionate BP125 mcg

Suspended in Propellant 134a.....q.s.

Seroflo-250 Inhaler (CFC FREE)

Each actuation delivers:

Salmeterol (as Salmeterol Xinafoate) Ph. Eur.....25 mcg

Fluticasone Propionate BP250 mcg

Suspended in Propellant 134a.....q.s.

Description

A one piece filled metal container fitted with a valve with plastic stem and printed with product name, batch number, and container no. On visual examination there is no sign of physical damage such as dents, bulged containers bent valve stem and uneven crimping.

Pharmacology

Pharmacodynamics

Pharmacotherapeutic Group: Adrenergics in combination with corticosteroids or other drugs, excl. Anticholinergics.

ATC Code: R03AK06

Asthma

Salmeterol Multi-center Asthma Research Trial (SMART)

The Salmeterol Multi-center Asthma Research Trial (SMART) was a 28-week US study that evaluated the safety of salmeterol compared to placebo added to usual therapy in adult and adolescent subjects. Although there were no significant differences in the primary endpoint of the combined number of respiratory-related deaths and respiratory-related life-threatening experiences, the study showed a

significant increase in asthma-related deaths in patients receiving salmeterol (13 deaths out of 13,176 patients treated with salmeterol versus 3 deaths out of 13,179 patients on placebo). The study was not designed to assess the impact of concurrent inhaled corticosteroid use.

Safety and efficacy of salmeterol-FP versus FP alone in asthma

Two multi-centre 26-week studies were conducted to compare the safety and efficacy of salmeterol-FP versus FP alone, one in adult and adolescent subjects (AUSTRI trial), and the other in paediatric subjects 4-11 years of age (VESTRI trial). For both studies, enrolled subjects had moderate to severe persistent asthma with history of asthma-related hospitalisation or asthma exacerbation in the previous year. The primary objective of each study was to determine whether the addition of LABA to ICS therapy (salmeterol-FP) was non-inferior to ICS (FP) alone in terms of the risk of serious asthma related events (asthma-related hospitalisation, endotracheal intubation, and death). A secondary efficacy objective of these studies was to evaluate whether ICS/LABA (salmeterol-FP) was superior to ICS therapy alone (FP) in terms of severe asthma exacerbation (defined as deterioration of asthma requiring the use of systemic corticosteroids for at least 3 days or an in-patient hospitalisation or emergency department visit due to asthma that required systemic corticosteroids).

A total of 11,679 and 6,208 subjects were randomized and received treatment in the AUSTRI and VESTRI trials, respectively. For the primary safety endpoint, non-inferiority was achieved for both trials (see Table below).

Serious Asthma-Related Events in the 26-Week AUSTRI and VESTRI Trials

	AUSTRI		VESTRI	
	Salmeterol-FP (n = 5,834)	FP Alone (n = 5,845)	Salmeterol-FP (n = 3,107)	FP Alone (n = 3,101)
Composite endpoint (Asthma-related hospitalisation, endotracheal intubation, -or death)	34 (0.6%)	33 (0.6%)	27 (0.9%)	21 (0.7%)
Salmeterol-FP/FP Hazard ratio (95% CI)	1.029 (0.638-1.662) ^a		1.285 (0.726-2.272) ^b	
Death	0	0	0	0
Asthma-related hospitalisation	34	33	27	21
Endotracheal intubation	0	2	0	0

^a If the resulting upper 95% CI estimate for the relative risk was less than 2.0, then non-inferiority was concluded.

^b If the resulting upper 95% CI estimate for the relative risk was less than 2.675, then non-inferiority was concluded.

For the secondary efficacy endpoint, reduction in time to first asthma exacerbation for salmeterol-FP relative to FP was seen in both studies, however only AUSTRI met statistical significance:

	AUSTRI		VESTRI	
	Salmeterol-FP (n = 5,834)	FP Alone (n = 5,845)	Salmeterol-FP (n = 3,107)	FP Alone (n = 3,101)
Number of subjects with an asthma exacerbation	480 (8%)	597 (10%)	265 (9%)	309 (10%)
Salmeterol-FP/FP Hazard ratio (95% CI)	0.787 (0.698, 0.888)		0.859 (0.729, 1.012)	

Twelve month study

A large twelve-month study (Gaining Optimal Asthma Control, GOAL) in 3416 asthma patients compared the efficacy and safety of Salmeterol and fluticasone propionate versus inhaled corticosteroid alone in achieving pre-defined levels of asthma control. Treatment was stepped-up every 12 weeks until ^{##}‘Total control’ was achieved or the highest dose of study drug was reached. Control needed to be sustained for at least 7 out of the last 8 weeks of treatment. The study showed that:

- 71% of patients treated with Salmeterol and fluticasone propionate achieved [#]‘Well-controlled’ asthma compared with 59% of patients treated with inhaled corticosteroid alone.
- 41% of patients treated with Salmeterol and fluticasone propionate achieved ^{##}‘Total control’ of asthma compared with 28% of patients treated with inhaled corticosteroid alone.

These effects were observed earlier with Salmeterol and fluticasone propionate compared with inhaled corticosteroid alone and at a lower inhaled corticosteroid dose.

The GOAL study also showed that:

- The rate of exacerbations was 29% lower with Salmeterol and fluticasone propionate compared to inhaled corticosteroid treatment alone.
- Attaining ‘Well controlled’ and ‘Totally controlled’ asthma improved Quality of Life (QoL). 61% of patients reported minimal or no impairment on QoL, as

measured by an asthma specific quality of life questionnaire, after treatment with Salmeterol and fluticasone propionate compared to 8% at baseline.

#Well controlled asthma; less than or equal to 2 days with symptom score greater than 1 (symptom score 1 defined as 'symptoms for one short period during the day'), SABA use on less than or equal to 2 days and less than or equal to 4 occasions/week, greater than or equal to 80% predicted morning peak expiratory flow, no night-time awakenings, no exacerbations and no side effects enforcing a change in therapy

##Total control of asthma; no symptoms, no SABA use, greater than or equal to 80% predicted morning peak expiratory flow, no night-time awakenings, no exacerbations and no side effects enforcing a change in therapy.

Two further studies have shown improvements in lung function, percentage of symptom free days and reduction in rescue medication use, at 60% lower inhaled corticosteroid dose with Salmeterol and fluticasone propionate compared to treatment with inhaled corticosteroid alone, whilst the control of the underlying airway inflammation, measured by bronchial biopsy and bronchoalveolar lavage, was maintained.

Additional studies have shown that treatment with Salmeterol and fluticasone propionate significantly improves asthma symptoms, lung function and reduces the use of rescue medication compared to treatment with the individual components alone and placebo. Results from GOAL show that the improvements seen with Salmeterol and fluticasone propionate, in these endpoints, are maintained over at least 12 months.

COPD

Symptomatic COPD patients without restriction to 10% reversibility to a short acting beta₂-agonist:-

Placebo-controlled clinical trials, over 6 months, have shown that regular use of both Salmeterol and fluticasone propionate 50/250 and 50/500 micrograms rapidly and significantly improves lung function, significantly reduced breathlessness and the use of relief medication. There were also significant improvements in health status.

Symptomatic COPD patients who demonstrated less than 10% reversibility to a short acting beta₂-agonist:-

Placebo-controlled clinical trials, over 6 and 12 months, have shown that regular use of Salmeterol and fluticasone propionate 50/500 micrograms rapidly and significantly improves lung function, significantly reduced breathlessness and the use of relief medication. Over a 12-month period the risk of COPD exacerbations and the need for additional courses of oral corticosteroids was significantly reduced. There were also significant improvements in health status.

Salmeterol and fluticasone propionate 50/500 micrograms was effective in improving lung function and health status and reducing the risk of COPD exacerbations, in both current and ex-smokers.

Fluticasone propionate containing medications in asthma during pregnancy

An observational retrospective epidemiological cohort study utilising electronic health records from the United Kingdom was conducted to evaluate the risk of MCMs following first trimester exposure to inhaled FP alone and Salmeterol and fluticasone propionate relative to non-FP containing ICS. No placebo comparator was included in this study.

Within the asthma cohort of 5362 first trimester ICS-exposed pregnancies, 131 diagnosed MCMs were identified; 1612 (30%) were exposed to FP or Salmeterol and fluticasone propionate of which 42 diagnosed MCMs were identified. The adjusted odds ratio for MCMs diagnosed by 1 year was 1.1 (95%CI: 0.5 – 2.3) for FP exposed vs non-FP ICS exposed women with moderate asthma and 1.2 (95%CI: 0.7 – 2.0) for women with considerable to severe asthma. No difference in the risk of MCMs was identified following first trimester exposure to FP alone versus Salmeterol and fluticasone propionate. Absolute risks of MCM across the asthma severity strata ranged from 2.0 to 2.9 per 100 FP-exposed pregnancies which is comparable to results from a study of 15,840 pregnancies unexposed to asthma therapies in the General Practice Research Database (2.8 MCM events per 100 pregnancies).

Mechanism of action

Seroflo Inhaler (CFC Free) contains salmeterol and fluticasone propionate which have differing modes of action. Salmeterol protects against symptoms, fluticasone propionate improves lung function and prevents exacerbations of the condition. Seroflo Inhaler (CFC Free) can offer a more convenient regime for patients on concurrent beta-agonist and inhaled corticosteroid therapy.

The respective mechanisms of action of both drugs are discussed below.

Salmeterol:

Salmeterol is a selective long-acting (12 hour) beta₂-adrenoceptor agonist with a long side chain which binds to the exo-site of the receptor.

Salmeterol produces a longer duration of bronchodilation, lasting for at least 12 hours, than recommended doses of conventional short-acting β₂ agonists.

These pharmacological properties of salmeterol offer more effective protection against histamine-induced bronchoconstriction and produce a longer duration of bronchodilation, lasting for at least 12 hours, than recommended doses of conventional short-acting beta₂-agonists.

In vitro tests have shown salmeterol is a potent and long-lasting inhibitor of the release, from human lung, of mast cell mediators such as histamine, leukotrienes and prostaglandin D₂.

In man salmeterol inhibits the early and late phase response to inhaled allergen; the latter persisting for over 30 hours after a single dose when the bronchodilator effect is no longer evident. Single dosing with salmeterol attenuates bronchial hyper-

responsiveness. These properties indicate that salmeterol has additional non-bronchodilator activity but the full clinical significance is not yet clear. This mechanism is different from the anti-inflammatory effect of corticosteroids.

Fluticasone propionate:

Fluticasone propionate given by inhalation at recommended doses has a glucocorticoid anti-inflammatory action within the lungs, resulting in reduced symptoms and exacerbations of asthma, without the adverse effects observed when corticosteroids are administered systemically.

Daily output of adrenocortical hormones usually remain within the normal range during chronic treatment with inhaled fluticasone propionate, even at the highest recommended doses in children and adults. After transfer from other inhaled steroids, the daily output gradually improves despite past and present intermittent use of oral steroids, thus demonstrating return of normal adrenal function on inhaled fluticasone propionate. The adrenal reserve also remains normal during chronic treatment, as measured by a normal increment on a stimulation test. However, any residual impairment of adrenal reserve from previous treatment may persist for a considerable time and should be borne in mind (see *Warnings and Precautions*).

Pharmacokinetics

There is no evidence in animals or human subjects that the administration of salmeterol and fluticasone propionate together by the inhaled route affects the pharmacokinetics of either component. For pharmacokinetic purposes therefore each component can be considered separately.

In a placebo-controlled, crossover drug interaction study in 15 healthy subjects, coadministration of Salmeterol (50 mcg twice daily inhaled) and the CYP3A4 inhibitor ketoconazole (400 mg once daily orally) for 7 days resulted in a significant increase in plasma salmeterol exposure (1.4- fold C_{max} and 15-fold AUC). There was no increase in salmeterol accumulation with repeat dosing. Three subjects were withdrawn from Salmeterol and ketoconazole co-administration due to QTc prolongation or palpitations with sinus tachycardia. In the remaining 12 subjects, coadministration of Salmeterol and ketoconazole did not result in a clinically significant effect on heart rate, blood potassium or QTc duration (see *Warnings and Precautions, and Interactions*).

Salmeterol:

Salmeterol acts locally in the lung therefore plasma levels are not an indication of therapeutic effects. In addition there are only limited data available on the pharmacokinetics of salmeterol because of the technical difficulty of assaying the drug in plasma due to the low plasma concentrations at therapeutic doses (approximately 200 picogram/ml or less) achieved after inhaled dosing. After regular dosing with salmeterol xinafoate, hydroxynaphthoic acid can be detected in the systemic circulation, reaching steady state concentrations of approximately 100 nanograms/mL. These concentrations are up to 1000 fold lower than steady state

levels observed in toxicity studies. No detrimental effects have been seen following long-term regular dosing (more than 12 months) in patients with airway obstruction.

An *in vitro* study showed that salmeterol is extensively metabolised to α -hydroxysalmeterol (aliphatic oxidation) by cytochrome P450 3A4 (CYP3A4). A repeat dose study with salmeterol and erythromycin in healthy volunteers showed no clinically significant changes in pharmacodynamic effects at 500 mg three times daily doses of erythromycin. However, a salmeterol-ketoconazole interaction study resulted in a significant increase in plasma salmeterol exposure (see *Warnings and Precautions, and Interactions*)

Fluticasone propionate:

The absolute bioavailability of fluticasone propionate for each of the available inhaler devices has been estimated from within and between study comparisons of inhaled and intravenous pharmacokinetic data. In healthy adult subjects the absolute bioavailability has been estimated for fluticasone propionate Accuhaler (7.8%), fluticasone propionate Diskhaler (9.0%), fluticasone propionate Evohaler (10.9%), salmeterol-fluticasone propionate Evohaler (5.3%) and salmeterol fluticasone propionate Accuhaler (5.5%) respectively. The absolute bioavailability of a single dose of inhaled fluticasone propionate in healthy subjects varies between approximately 5 to 11% of the nominal dose depending on the inhalation device used. In patients with asthma or COPD a lesser degree of systemic exposure to inhaled fluticasone propionate has been observed.

Systemic absorption occurs mainly through the lungs and is initially rapid then prolonged. The remainder of the inhaled dose may be swallowed but contributes minimally to systemic exposure due to the low aqueous solubility and pre-systemic metabolism, resulting in oral availability of less than 1%. There is a linear increase in systemic exposure with increasing inhaled dose.

The disposition of fluticasone propionate is characterized by high plasma clearance (1150ml/min), a large volume of distribution at steady-state (approximately 300 L) and a terminal half-life of approximately 8 hours.

Plasma protein binding is moderately high (91%). Fluticasone propionate is cleared very rapidly from the systemic circulation, principally by metabolism to an inactive carboxylic acid metabolite, by the cytochrome P450 enzyme CYP3A4. Other unidentified metabolites are also found in the faeces.

The renal clearance of fluticasone propionate is negligible (<0.2%) and less than 5% as the metabolite. Care should be taken when co-administering known CYP3A4 inhibitors, as there is potential for increased systemic exposure to fluticasone propionate. The main part of the dose is excreted as faeces as metabolites and unchanged drug.

Non-Clinical Information

Salmeterol xinafoate and fluticasone propionate have been extensively evaluated in animal toxicity tests. Significant toxicities occurred only at doses in excess of those recommended for human use and were those expected for a potent beta₂-

adrenoreceptor agonist and glucocorticosteroid. Neither salmeterol xinafoate or fluticasone propionate have shown any potential for genetic toxicity.

In long term studies, salmeterol xinafoate induced benign tumours of smooth muscle in the mesovarium of rats and the uterus of mice.

Rodents are sensitive to the formation of these pharmacologically-induced tumours. Salmeterol is not considered to represent a significant oncogenic hazard to man.

Co-administration of salmeterol and fluticasone propionate resulted in some cardiovascular interactions at high doses. In rats, mild atrial myocarditis and focal coronary arteritis were transient effects that resolved with regular dosing. In dogs, heart rate increases were greater after co-administration than after salmeterol alone. No clinically relevant serious adverse cardiac effects have been observed in studies in man.

Co-administration did not modify other class-related toxicities in animals.

The non-CFC propellant, HFA134a, has been shown to have no toxic effect at very high vapour concentrations, far in excess of those likely to be experienced by patients, in a wide range of animal species exposed daily for periods of two years.

Indications

Asthma (Reversible Obstructive Airways Disease)

Seroflo Inhaler (CFC Free) is indicated in the regular treatment of reversible obstructive airways disease (ROAD), including asthma in children and adults, where use of a combination (bronchodilator and inhaled corticosteroid) is appropriate.

This may include:

Patients on effective maintenance doses of long-acting beta-agonists and inhaled corticosteroids.

Patients who are symptomatic on current inhaled corticosteroid therapy.

Patients on regular bronchodilator therapy who require inhaled corticosteroids.

Chronic Obstructive Pulmonary Disease (COPD)

Seroflo Inhaler (CFC Free) is indicated for the regular treatment of chronic obstructive pulmonary disease (COPD) including chronic bronchitis and emphysema.

Dosage and Administration

Seroflo Inhaler (CFC Free) is for inhalation only.

Patients should be made aware that Seroflo Inhaler (CFC Free) must be used regularly for optimum benefit, even when asymptomatic.

Patients should be regularly reassessed by a doctor, so that the strength of Seroflo Inhaler (CFC Free) they are receiving remains optimal and is only changed on medical advice.

Asthma (Reversible Obstructive Airways Disease)

The dose should be titrated to the lowest dose at which effective control of symptoms is maintained.

Where the control of symptoms is maintained with twice daily salmeterol and fluticasone propionate, titration to the lowest effective dose could include salmeterol and fluticasone propionate given once daily.

Patients should be given the strength of Seroflo Inhaler (CFC Free) containing the appropriate fluticasone propionate dosage for the severity of their disease.

If a patient is inadequately controlled on inhaled corticosteroid therapy alone, substitution with salmeterol and fluticasone propionate at a therapeutically equivalent corticosteroid dose may result in an improvement in asthma control. For patients whose asthma control is acceptable on inhaled corticosteroid therapy alone, substitution with salmeterol and fluticasone propionate may permit a reduction in corticosteroid dose while maintaining asthma control. For further information, please refer to the 'Pharmacodynamics' section.

Recommended Doses:

Adults and adolescents 12 years and older:

Two inhalations of 25 micrograms salmeterol and 50 micrograms fluticasone propionate twice daily.

or

Two inhalations of 25 micrograms salmeterol and 125 micrograms fluticasone propionate twice daily.

or

Two inhalations of 25 micrograms salmeterol and 250 micrograms fluticasone propionate twice daily.

Children 4 years and older:

Two inhalations of 25 micrograms salmeterol and 50 micrograms fluticasone propionate twice daily.

There are no data available for use of salmeterol and fluticasone propionate in children aged under 4 years.

Chronic Obstructive Pulmonary Disease (COPD):

For adult patients the recommended dose is two inhalations 25/125 micrograms to 25/250 micrograms salmeterol/fluticasone propionate twice daily.

Special patient groups:

There is no need to adjust the dose in elderly patients or in those with renal or hepatic impairment.

Mode of administration

Oral Inhalation

Contraindications

Salmeterol/fluticasone propionate is contraindicated in patients with a history of hypersensitivity to any of the ingredients.

Warnings and Precautions

Salmeterol/fluticasone propionate inhaler is not for relief of acute symptoms for which a fast and short-acting bronchodilator (e.g. salbutamol) is required. Patients should be advised to have their relief medication available at all times.

Patients should not be initiated on salmeterol/fluticasone propionate during an exacerbation, or if they have significantly worsening or acutely deteriorating asthma.

Serious asthma-related adverse events and exacerbations may occur during treatment with salmeterol/fluticasone propionate. Patients should be asked to continue treatment but to seek medical advice if asthma symptoms remain uncontrolled or worsen after initiation on salmeterol/fluticasone propionate.

Increasing use of short-acting bronchodilators to relieve symptoms indicates deterioration of control and patients should be reviewed by a physician.

Sudden and progressive deterioration in control of asthma is potentially life-threatening and the patient should be reviewed by a physician. Consideration should be given to increasing corticosteroid therapy. Also, where the current dosage of Salmeterol/fluticasone propionate has failed to give adequate control of asthma, the patient should be reviewed by a physician.

Salmeterol/fluticasone propionate should not be initiated in patients with unstable and acutely deteriorating asthma, which may be a life-threatening condition. Serious acute respiratory events, including fatalities, have been reported when salmeterol has been initiated in this situation. Although it is not possible from these reports to determine whether salmeterol contributed to these adverse events or failed to relieve the deteriorating asthma, the use of salmeterol in this setting is inappropriate.

Once asthma symptoms are controlled, consideration may be given to gradually reducing the dose of salmeterol/fluticasone propionate. Regular review of patients as treatment is stepped down is important. The lowest effective dose of salmeterol/fluticasone propionate should be used (see *Dosage and Administration*).

Treatment with salmeterol/fluticasone propionate should not be stopped abruptly in patients with asthma due to risk of exacerbation, therapy should be down-titrated under physician supervision. For patients with COPD cessation of therapy may be associated with symptomatic decompensation and should be supervised by a physician.

There was an increased reporting of pneumonia in studies of patients with COPD receiving Salmeterol / fluticasone propionate (see *Adverse Reactions*). Physicians should remain vigilant for the possible development of pneumonia in patients with COPD as the clinical features of pneumonia and exacerbation frequently overlap.

As with all inhaled medication containing corticosteroids, salmeterol/fluticasone propionate should be administered with caution in patients with active or quiescent pulmonary tuberculosis and fungal, viral or other infections of the airway. Appropriate treatment should be promptly instituted, if indicated.

Salmeterol/fluticasone propionate should be administered with caution in patients with thyrotoxicosis.

Cardiovascular effects, such as increases in systolic blood pressure and heart rate, may occasionally be seen with all sympathomimetic drugs, especially at higher than therapeutic doses. For this reason, Salmeterol/fluticasone propionate should be used with caution in patients with pre-existing cardiovascular disease.

A transient decrease in serum potassium may occur with all sympathomimetic drugs at higher therapeutic doses. Therefore, Salmeterol/fluticasone propionate should be used with caution in patients predisposed to low levels of serum potassium.

Systemic effects may occur with any inhaled corticosteroid, particularly at high doses prescribed for long periods; these effects are much less likely to occur than with oral corticosteroids (see *Overdose*). Possible systemic effects include Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract, glaucoma and central serous chorioretinopathy. More rarely, a range of psychological or behavioural effects including psychomotor hyperactivity, sleep disorders, anxiety, depression or aggression (particularly in children) (see Paediatric population sub-heading below for information on the systemic effects of inhaled corticosteroids in children and adolescents).

It is important, therefore for asthma patients, that the dose of inhaled corticosteroid is titrated to the lowest dose at which effective control is maintained.

The possibility of impaired adrenal response should always be borne in mind in emergency and elective situations likely to produce stress and appropriate corticosteroid treatment considered (see *Overdose*).

It is recommended that the height of children receiving prolonged treatment with inhaled corticosteroid is regularly monitored.

Because of the possibility of impaired adrenal response, patients transferring from oral steroid therapy to inhaled fluticasone propionate therapy should be treated with special care, and adrenocortical function regularly monitored.

There have been very rare reports of increases in blood glucose levels (see *Adverse Reactions*) and this should be considered when prescribing to patients with a history of diabetes mellitus.

During post-marketing use, there have been reports of clinically significant drug interactions in patients receiving fluticasone propionate and ritonavir, resulting in systemic corticosteroid effects including Cushing's syndrome and adrenal suppression. Therefore, concomitant use of fluticasone propionate and ritonavir should be avoided, unless the potential benefit to the patient outweighs the risk of systemic corticosteroid side-effects (see *Interactions*).

It was observed in a drug interaction study that concomitant use of systemic ketoconazole increases exposure to Salmeterol/fluticasone propionate. This may lead to prolongation in the QTc interval. Caution should be exercised when strong CYP3A4 inhibitors (e.g. ketoconazole) are co-administered with Salmeterol (see *Interactions, and Pharmacokinetics*).

As with other inhalation therapy paradoxical bronchospasm may occur with an immediate increase in wheezing after dosing. This should be treated immediately with a fast and shortacting inhaled bronchodilator. Salmeterol/fluticasone inhaler should be discontinued immediately, the patient assessed and alternative therapy instituted if necessary (see *Adverse Reactions*).

The pharmacological side-effects of beta-2 agonist treatment, such as tremor, subjective palpitations and headache have been reported, but tend to be transient and to reduce with regular therapy (see *Adverse Reactions*).

Prolonged treatment of patients with high doses of inhaled corticosteroids may result in adrenal suppression and acute adrenal crisis. Very rare cases of adrenal suppression and acute adrenal crisis have also been described with doses of fluticasone propionate between 500 and less than 1000 micrograms. Situations, which could potentially trigger acute adrenal crisis, include trauma, surgery, infection or any rapid reduction in dosage. Presenting symptoms are typically vague and may include anorexia, abdominal pain, weight loss, tiredness, headache, nausea, vomiting, hypotension, decreased level of consciousness, hypoglycaemia, and seizures. Additional systemic corticosteroid cover should be considered during periods of stress or elective surgery.

Systemic absorption of salmeterol and fluticasone propionate is largely through the lungs. As the use of a spacer device with a metered dose inhaler may increase drug delivery to the lungs it should be noted that this could potentially lead to an increase in the risk of systemic adverse effects. Single dose pharmacokinetic data have demonstrated that the systemic exposure to salmeterol and fluticasone propionate

may be increased as much as two-fold when the spacer device is used with salmeterol xinafoate and fluticasone propionate inhaler as compared with the spacer device.

Patients who have required high dose emergency corticosteroid therapy in the past may also be at risk. The extent of the adrenal impairment may require specialist advice before elective procedures.

Pneumonia in patients with COPD

An increase in the incidence of pneumonia, including pneumonia requiring hospitalisation, has been observed in patients with COPD receiving inhaled corticosteroids. There is some evidence of an increased risk of pneumonia with increasing steroid dose but this has not been demonstrated conclusively across all studies.

There is no conclusive clinical evidence for intra-class differences in the magnitude of the pneumonia risk among inhaled corticosteroid products.

Physicians should remain vigilant for the possible development of pneumonia in patients with COPD as the clinical features of such infections overlap with the symptoms of COPD exacerbations.

Risk factors for pneumonia in patients with COPD include current smoking status, older age, low body mass index (BMI) and severe COPD.

Ritonavir can greatly increase the concentration of fluticasone propionate in plasma. Therefore, concomitant use should be avoided, unless the potential benefit to the patient outweighs the risk of systemic corticosteroid side effects. There is also an increased risk of systemic side effects when combining fluticasone propionate with other potent CYP3A inhibitors (see Drug interactions).

There was an increased reporting of lower respiratory tract infections (particularly pneumonia and bronchitis) in a 3-year study in patients with Chronic Obstructive Pulmonary Disease (COPD) receiving salmeterol /fluticasone propionate as a fixed-dose combination administered via the dry powder inhaler compared with placebo (see Undesirable Effects). In a 3 year COPD study, older patients, patients with a lower body mass index ($<25\text{kg/m}^2$) and patients with very severe disease ($\text{FEV}_1 < 30\%$ predicted) were at greatest risk of developing pneumonia regardless of treatment. If a patient with severe COPD has experienced pneumonia the treatment with salmeterol/fluticasone propionate should be re-evaluated.

Paediatric population

Children and adolescents <16 years taking high doses of fluticasone propionate (typically ≥ 1000 micrograms/day) may be at particular risk of systemic effects. Systemic effects may occur, particularly at high doses prescribed for long periods. Possible systemic effects include Cushing's syndrome, Cushingoid features, adrenal suppression, acute adrenal crisis and growth retardation in children and adolescents

and more rarely, a range of psychological or behavioural effects including psychomotor hyperactivity, sleep disorders, anxiety, depression or aggression. Consideration should be given to referring the child or adolescent to a paediatric respiratory specialist.

The dose of inhaled corticosteroid should be reduced to the lowest dose at which effective control of asthma is maintained.

Interactions

Both non-selective and selective beta-blockers should be avoided unless there are compelling reasons for their use.

Concomitant use of other beta-adrenergic containing drugs can have a potentially additive effect.

Fluticasone Propionate

Under normal circumstances, low plasma concentrations of fluticasone propionate are achieved after inhaled dosing, due to extensive first pass metabolism and high systemic clearance mediated by cytochrome P450 3A4 in the gut and liver. Hence, clinically significant drug interactions mediated by fluticasone propionate are unlikely.

A drug interaction study in healthy subjects has shown that ritonavir (a highly potent cytochrome P450 3A4 inhibitor) can greatly increase fluticasone propionate plasma concentrations, resulting in markedly reduced serum cortisol concentrations. During post-marketing use, there have been reports of clinically significant drug interactions in patients receiving intranasal or inhaled fluticasone propionate and ritonavir, resulting in systemic corticosteroid effects including Cushing's syndrome and adrenal suppression. Therefore, concomitant use of fluticasone propionate and ritonavir should be avoided, unless the potential benefit to the patient outweighs the risk of systemic corticosteroid side-effects.

Studies have shown that other inhibitors of cytochrome P450 3A4 produce negligible (erythromycin) and minor (ketoconazole) increases in systemic exposure to fluticasone propionate without notable reductions in serum cortisol concentrations. Nevertheless, care is advised when co-administering potent cytochrome P450 3A4 inhibitors (e.g. ketoconazole) as there is potential for increased systemic exposure to fluticasone propionate.

Co-administration of ketoconazole and Salmeterol/fluticasone propionate resulted in a significant increase in plasma salmeterol exposure (1.4-fold C_{max} and 15-fold AUC) and this may cause a prolongation of the QTc interval (see *Warnings and Precautions, and Pharmacokinetics*).

In a small study in healthy volunteers, the slightly less potent CYP3A inhibitor ketoconazole increased the exposure of fluticasone propionate after a single inhalation by 150%. This resulted in a greater reduction of plasma cortisol as compared with fluticasone propionate alone. Co-treatment with other potent CYP3A inhibitors, such as itraconazole, is also expected to increase the systemic fluticasone

propionate exposure and the risk of systemic side-effects. Caution is recommended and long-term treatment with such drugs should if possible be avoided.

Salmeterol

Potent CYP3A4 inhibitors

Co-administration of ketoconazole (400 mg orally once daily) and salmeterol (50 micrograms inhaled twice daily) in 15 healthy subjects for 7 days resulted in a significant increase in plasma salmeterol exposure (1.4-fold C_{max} and 15-fold AUC). This may lead to an increase in the incidence of other systemic effects of salmeterol treatment (e.g. prolongation of QTc interval and palpitations) compared with salmeterol or ketoconazole treatment alone (see Section Warnings and Precautions).

Clinically significant effects were not seen on blood pressure, heart rate, blood glucose and blood potassium levels. Co-administration with ketoconazole did not increase the elimination half-life of salmeterol or increase salmeterol accumulation with repeat dosing.

The concomitant administration of ketoconazole should be avoided, unless the benefits outweigh the potentially increased risk of systemic side effects of salmeterol treatment. There is likely to be a similar risk of interaction with other potent CYP3A4 inhibitors (e.g. itraconazole, telithromycin, ritonavir).

Moderate CYP 3A4 inhibitors

Co-administration of erythromycin (500mg orally three times a day) and salmeterol (50 micrograms inhaled twice daily) in 15 healthy subjects for 6 days resulted in a small but non-statistically significant increase in salmeterol exposure (1.4-fold C_{max} and 1.2-fold AUC). Co-administration with erythromycin was not associated with any serious adverse effects.

Pregnancy and lactation

There are no data on human fertility. Animal studies indicate no effects of salmeterol or Fluticasone propionate on male or female fertility.

There are limited data in pregnant women. Administration of drugs during pregnancy should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus or child.

Results from a retrospective epidemiological study did not find an increased risk of major congenital malformations (MCMs) following exposure to fluticasone propionate when compared to other inhaled corticosteroids, during the first trimester of pregnancy (see *Pharmacodynamics*).

Reproductive toxicity studies in animals, either with single drug or in combination, revealed the foetal effects expected at excessive systemic exposure levels of a potent beta₂-adrenoreceptor agonist and glucocorticosteroid. Extensive clinical experience with drugs in these classes has revealed no evidence that the effects are relevant at therapeutic doses.

The lowest effective dose of fluticasone propionate needed to maintain adequate asthma control should be used in the treatment of pregnant women.

Salmeterol and fluticasone propionate concentrations in plasma after inhaled therapeutic doses are very low and therefore concentrations in human breast milk are likely to be correspondingly low. This is supported by studies in lactating animals, in which low drug concentrations were measured in milk. There are no data available for human breast milk.

Administration during lactation should only be considered if the expected benefit to the mother is greater than any possible risk to the child.

Effects on Ability to Drive and Use Machines

There have been no specific studies of the effect of Salmeterol / fluticasone propionate on the above activities, but the pharmacology of both drugs does not indicate any effect.

Adverse Reactions

All of the adverse reactions associated with the individual components, salmeterol and fluticasone propionate, are listed below. There are no additional adverse reactions attributed to the combination product when compared to the adverse event profiles of the individual components.

Adverse events which have been associated with salmeterol/fluticasone propionate are given below, listed by system organ class and frequency. Frequencies are defined as: very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1000$ to $< 1/100$), rare ($\geq 1/10,000$ to $< 1/1000$) and very rare ($< 1/10,000$). The majority of frequencies were determined from pooled clinical trial data from 23 asthma and 7 COPD studies. Not all events were reported in clinical trials. For these events, the frequency was calculated based on spontaneous data.

System Organ Class	Adverse Event	Frequency
Infections and infestations	Candidiasis of the mouth and throat	Common
	Pneumonia	Common ^{1,3}
	Bronchitis	Common ^{1,3}
	Oesophageal candidiasis	Rare
Immune system disorders	Hypersensitivity Reactions: manifesting as:	
	Cutaneous hypersensitivity reactions	Uncommon
	Angioedema (mainly facial and oropharyngeal oedema)	
	dyspnoea	Rare

	Respiratory symptoms (bronchospasm) Anaphylactic reactions including anaphylactic shock	Uncommon Rare Rare
Endocrine Disorders	Possible systemic effects include (see <i>Warnings and Precautions</i>) Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation in children and adolescents, decreased bone mineral density	Rare ⁴
Metabolism and nutrition disorders	Hypokalaemia Hyperglycaemia	Common ³ Uncommon ⁴
Psychiatric disorders	Anxiety Sleep disorders Behavioural changes, including hyperactivity and irritability (predominantly in children) Depression, aggression (predominantly in children)	Uncommon Uncommon Rare Not Known
Nervous system disorders	Headache (see <i>Warnings and Precautions</i>) Tremor (see <i>Warnings and Precautions</i>)	Very Common ¹ Uncommon
Eye disorder	Cataract Glaucoma	Uncommon Rare ⁴
Cardiac disorders	Palpitations (see <i>Warnings and Precautions</i>) Tachycardia Cardiac arrhythmias including supraventricular tachycardia and extrasystoles. Atrial fibrillation Angina pectoris	Uncommon Uncommon Rare Uncommon Uncommon
Respiratory, thoracic and mediastinal disorders	Nasopharyngitis Throat irritation Hoarseness/dysphonia Sinusitis Paradoxical bronchospasm (see <i>Warnings and Precautions</i>)	Very Common ^{2,3} Uncommon Common Common ^{1,3} Rare ⁴
Skin and subcutaneous tissue disorders	Contusions	Uncommon ^{1,3}
Musculoskeletal and connective tissue disorders	Muscle cramps Traumatic fractures Arthralgia Myalgia	Common Common ^{1,3} Common Common

1. Reported commonly in placebo
2. Reported very commonly in placebo
3. Reported over 3 years in a COPD study
4. See Warnings and Precautions

Description of selected adverse reactions

The pharmacological side effects of β_2 agonist treatment, such as tremor, palpitations and headache, have been reported, but tend to be transient and reduce with regular therapy.

As with other inhalation therapy paradoxical bronchospasm may occur with an immediate increase in wheezing and shortness of breath after dosing. Paradoxical bronchospasm responds to a rapid-acting bronchodilator and should be treated straightaway. Salmeterol and fluticasone propionate inhaler should be discontinued immediately, the patient assessed and alternative therapy instituted if necessary.

Due to the fluticasone propionate component, hoarseness and candidiasis (thrush) of the mouth and throat and, rarely, of the oesophagus can occur in some patients. Both hoarseness and incidence of mouth and throat candidiasis may be relieved by rinsing the mouth with water and/or brushing the teeth after using the product. Symptomatic mouth and throat candidiasis can be treated with topical anti-fungal therapy whilst still continuing with the Salmeterol and fluticasone propionate inhaler.

Paediatric population

Possible systemic effects include Cushing's syndrome, Cushingoid features, adrenal suppression and growth retardation in children and adolescents (see Warnings and Precautions). Children may also experience anxiety, sleep disorders and behavioural changes, including hyperactivity and irritability.

Overdose

The available information on overdose with Salmeterol / fluticasone propionate is given below:

The expected symptoms and signs of salmeterol overdosage are those of excessive β_2 -adrenergic stimulation, including tremor, headache, tachycardia, increase in systolic blood pressure and hypokalaemia. There is no specific treatment for an overdose of salmeterol and fluticasone propionate. If overdose occurs, the patients should be treated supportively with appropriate monitoring as necessary. The preferred antidotes are cardioselective beta-blocking agents, which should be used with caution in patients with a history of bronchospasm. If salmeterol/fluticasone propionate therapy has to be withdrawn due to overdose of the beta agonist component of the drug, provision of appropriate replacement steroid therapy should be considered. Additionally, hypokalaemia can occur and potassium replacement should be considered.

Acute inhalation of fluticasone propionate doses in excess of those approved may lead to temporary suppression of the hypothalamic-pituitary-adrenal axis. This does not

usually require emergency action as normal adrenal function typically recovers within a few days.

If higher than approved doses of Salmeterol / fluticasone propionate are continued over prolonged periods, significant adrenocortical suppression is possible. There have been very rare reports of acute adrenal crisis, mainly occurring in children exposed to higher than approved doses over prolonged periods (several months or years); observed features have included hypoglycaemia associated with decreased consciousness and/or convulsions. Situations which could potentially trigger acute adrenal crisis include exposure to trauma, surgery, infection or any rapid reduction in the dosage of the inhaled fluticasone propionate component.

It is not recommended that patients receive higher than approved doses of Salmeterol / fluticasone propionate. It is important to review therapy regularly and titrate down to the lowest approved dose at which effective control of disease is maintained (see *Dosage and Administration*).

Chronic overdose of inhaled fluticasone propionate: Refer to section *Warnings and Precautions*: risk of adrenal suppression.

Monitoring of adrenal reserve may be necessary. In cases of fluticasone propionate overdose salmeterol/fluticasone propionate therapy may still be continued at a suitable dosage for symptom control.

Storage and Handling Instructions

Store below 30°C. Do not freeze.

Shelf- Life: 24 months

Pack Size: Carton containing Canister of 120 metered doses.

Registration Holder in Malaysia

Cipla Malaysia SDN BHD
Suite 1101, Amcorp Tower,
Amcorp Trade Centre, 18 Persiaran Barat,
46050 Petaling Jaya,
Selangor, Malaysia

Manufactured By

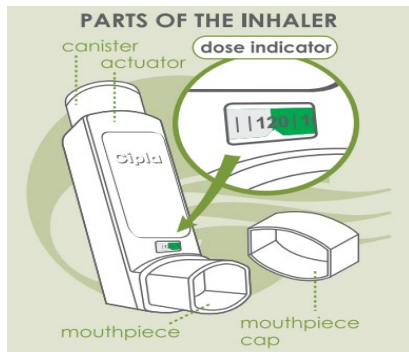
Cipla Ltd.
L-139 to L-146
Verna Industrial Estate
Verna Goa, India

Date of Revision

Dec 2025

Patient Information

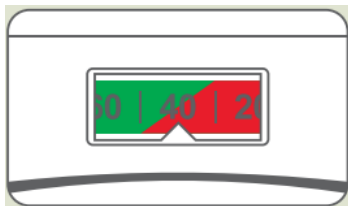
About your Seroflo Inhaler



Your Seroflo inhaler now comes with a dose indicator. It shows the number of puffs in the inhaler. As you use the inhaler, the dose indicator will countdown and indicate the number of puffs remaining.

How to know that your Seroflo Inhaler is getting over

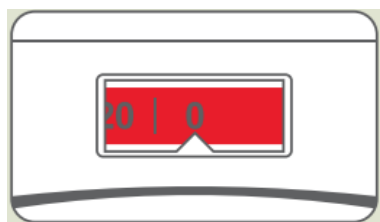
When there are 40 puffs remaining, the colour on the dose indicator will change from green to red.



This indicates that fewer doses are remaining in the inhaler. You should now consider getting a new inhaler or ask your doctor if you need another one.

When the dose indicator displays '0',

This means that there is more medicine left in the inhaler & you need to discard the inhaler. Your inhaler may not feel empty & it may continue to operate, but you will not get the right amount of medicine, if you keep using it beyond '0'.



Instructions for use

Testing your inhaler Before using for the first time, remove the mouthpiece cover by gently squeezing the sides of the cover, shake the inhaler well, hold the inhaler between fingers and thumb with your thumb at the base, below the mouthpiece and release puffs into the air until the counter reads 120 to make sure that it works. The inhaler should be shaken immediately before releasing each puff. If your inhaler has

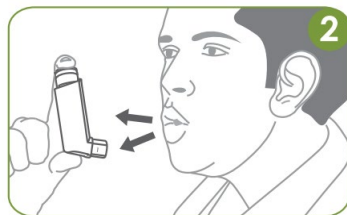
not been used for a week or more remove the mouthpiece cover, shake the inhaler well and release one puff into the air. Each time the inhaler is activated the number on the counter will count down by one. In certain circumstances, dropping the inhaler may cause the counter to count on.

Using your inhaler

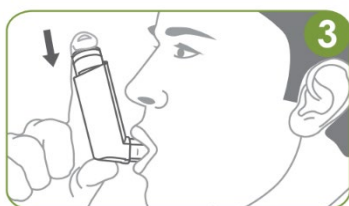
1. *Sit or stand upright. Remove the mouthpiece cap & shake the inhaler well. Hold it upright as shown, with your thumb at the base below the mouthpiece. Place either one or two fingers on top of the canister.*



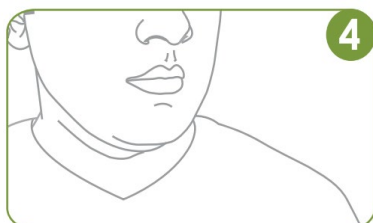
2. *Breathe out fully, through your mouth.*



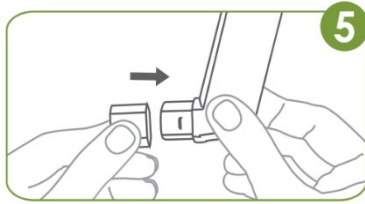
3. *Place the mouthpiece of the inhaler in your mouth between your teeth & close your lips around it (do not bite it). Start breathing in slowly through your mouth. Press down the canister firmly & fully to release one spray while continuing to breathe in slowly & deeply.*



4. *Remove the inhaler from your mouth & hold your breath for 10 seconds, or for as long as is comfortable. Breathe out normally.*



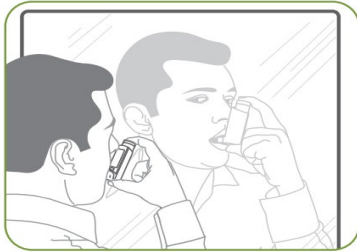
5. *If another puff is required, wait for at least 1 minute. Shake inhaler well & repeat step 2 to 4. After use, replace the mouthpiece cap firmly & snap it into position.*



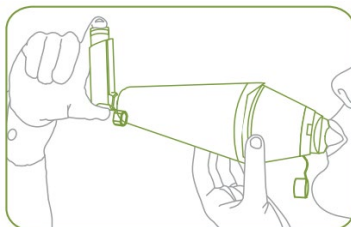
6. *After taking each dose, rinse your mouth with water & spit it out.*

Do not rush steps 2, 3 & 4. It is important that you start to breathe in slowly before releasing a puff.

To ensure correct use of the inhaler, use it in front of a mirror for the first few times. If you see 'mist' coming out from the top of the inhaler or the sides of your mouth, start again from step 1. This escaping mist indicates incorrect technique.



In case of difficulty in using the inhaler correctly, you may use it along with a spacer device.



Children

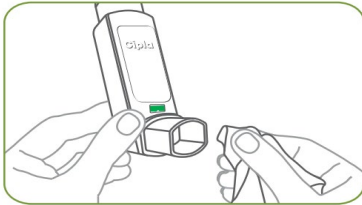
Parents must assist those children who need help in using the SEROFLO inhaler correctly with/without a spacer.



Cleaning your Seroflo Inhaler

It is important to keep your inhaler clean. Clean your inhaler atleast once a week.

1. Take the mouthpiece cap off. **DO NOT take the metal canister out of the actuator.**
2. Wipe the inside & the outside of the mouthpiece with a clean, dry cloth.



3. Replace the mouthpiece cap.
4. **DO NOT wash or soak any part of the inhaler in water.**

Storage and Disposal of Seroflo

- Storage

Store below 30°C.

Do not freeze.

Keep the inhaler in an upright position, with the mouthpiece down.

DO NOT

- **Spray the inhaler in your eyes.**
- **Exceed the recommended dose.**
- **Change/tamper with the numbers on the dose indicator.**
- **Puncture or burn the inhaler even when empty as it is pressurized.**
- **Keep the inhaler out of the reach of children.**

Disposal

- Medicines should not be disposed of via wastewater or household waste. Ask your pharmacist how to dispose of medicines no longer required. These measures will help to protect the environment.